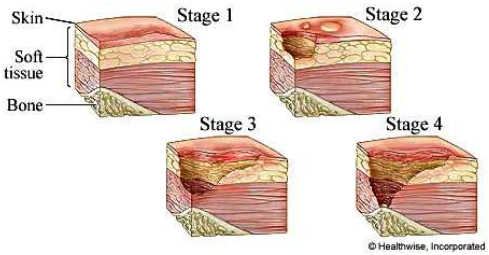
Evaluating Techniques for Pressure Wound Classification Using ChatGPT

Kobe Richards (Data Science)
Advisor: Professor Emmanuel Agu (Computer Science)
PhD Student Advisor: Reza Saadati Fard (Computer Science)

Introduction

- Accurate wound classification is crucial for effective treatment
- Provider training differences and overlapping wound characteristics can lead to variability and misclassification
- This project focuses on the Pressure Injury Images Dataset (PIID) containing 1091 pressure injury images and their classifications
- Pressure injuries have 4 stages, ranging from mild redness to deep wounds exposing muscle or bone

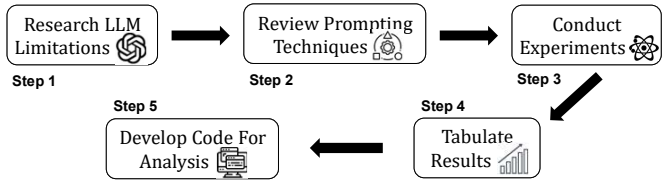


[1] **Figure 1: Diagram showing the variation of the 4 stages of pressure injuries.**

Research Objective

To evaluate and compare the effectiveness of various prompt engineering techniques, using ChatGPT, in accurately classifying the severity of pressure wounds in medical images.

Methodology



Prompting Techniques:

[3] CO-STAR: Structures the input in a context-objective-style-tone-audience-response format to guide the model's output.

Zero-Shot: Asking the model to perform a task it has not been explicitly trained on, relying solely on its pre-existing knowledge and understanding.

Few-Shot: Providing the model with a few examples of the task to guide its response, improving accuracy and contextual understanding.

[3] TIDD-EC: Directs the model to follow specific instruction sin a task-instructions-do's-don'ts-examples-content format to generate accurate responses.

Chain-of-Thought: Encourages the model to think trough the problem step-by-step, providing intermediate reasoning steps before arriving at the final answer.

Tree-of-Thought: Multiple perspectives (i.e., medical professionals) are considered sequentially or in parallel, each contributing to the final decision or classification.

Technique	Unweighted Accuracy (± Std Dev)	Weighted Accuracy (± Std Dev)	Precision	Recall	F1 Score
CO-STAR	0.700 ± 0.232	0.869 ± 0.114	0.755	0.700	0.690
Zero-Shot	0.700 ± 0.224	0.872 ± 0.109	0.755	0.700	0.684
Few-Shot	0.537 ± 0.303	0.805 ± 0.150	0.633	0.537	0.499
TIDDEC	0.600 ± 0.281	0.820 ± 0.152	0.695	0.600	0.569
Chain-of-Thought	**0.725 ± 0.144**	0.887 ± 0.093	**0.799**	**0.725**	**0.728**
Tree-of-Thought	0.662 ± 0.204	**0.887 ± 0.066**	0.684	0.662	0.651

Table 1: Quantitative overview of performance metrics for each prompting technique. The overall best and worst performers are highlighted in green and red, respectively. The best performing technique for each column is emboldened in green text.

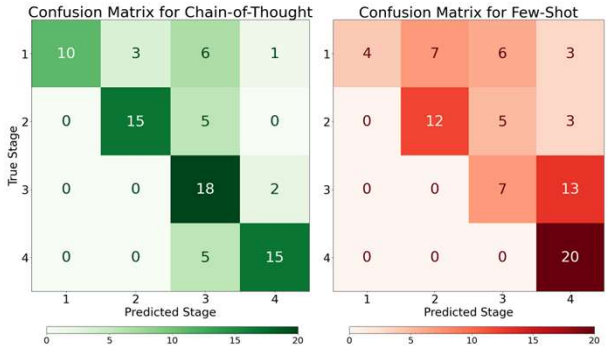


Figure 2: Confusion matrices for the best and worst performing prompting techniques, illustrating their predictions across wound stages and their ability to differentiate between classes.

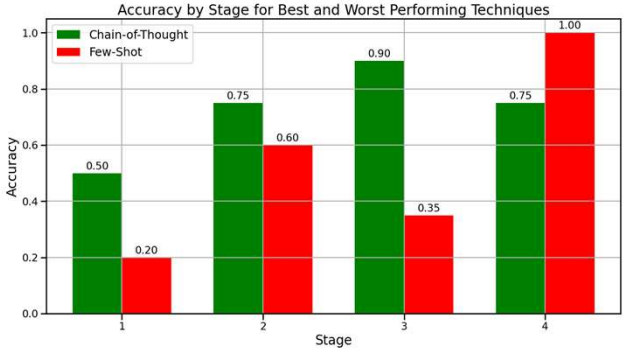


Figure 3: Histogram of accuracy by stage for the best and worst performing prompting techniques, displaying their performance variations across different wound stages.

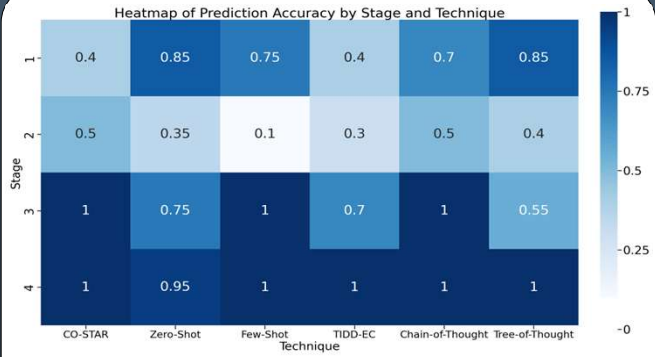


Figure 4: This heatmap shows the consistency of accurate predictions of each technique across wound stages, based on ten repetitions for two images per stage. Darker shades represent higher accuracy, lighter shades indicate lower accuracy.

Discussion

- Improved accuracy with the specification of the use of the PIID classification scale in prompts
- Techniques like Chain-of-Thought and CO-STAR, which required minimal tuning, were more effective than those needing detailed examples and more sophisticated approaches
- Limited to 20 images per stage for main analysis due to time constraints
- High accuracy across stages in the heatmap suggests reliable reproducibility, but some prompts need more testing and adjustments

Next Steps

- Analyze other medical image datasets (e.g., AZH) for different classification problems
- Automate input of images and prompting techniques in ChatGPT using APIs
- Integrate findings and AI technologies to develop a mobile application to aid medical professionals in wound diagnosis

Acknowledgments

NSF REU grant EEC2150076

SmartWANDS Group

References

[1] Healthwise, Incorporated. (n.d.). Stages of pressure injuries. Cigna. Retrieved July 22, 2024, from https://www.cigna.com/knowledge-center/hw/stages-of-pressure-injuries-zm2442

[2] OpenAI. (2023). *GPT-4 technical report*. arXiv. https://arxiv.org/pdf/2303.08774

[3] Vivas.ai. (n.d.). *Mastering prompt engineering: A guide to the CO-STAR and TIDD-EC frameworks*. Vivas.ai. Retrieved July 23, 2024, from https://vivas.ai/blogs/mastering-prompt-engineering-a-guide-to-the-co-star-and-tidd-ec-frameworks

[4] FU-MedicalAI. (n.d.). *PIID* [GitHub repository]. Retrieved July 24, 2024, from https://github.com/FU-MedicalAI/PIID?tab=readme-ov-file